

MYELOID NEOPLASIA

Aberrant splicing of *MBD1* reshapes the epigenome to drive convergent myeloerythroid defects in MDS

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KEY POINTS

- Global missplicing of *MBD1* represents a novel gain-of-function epigenetic axis driving erythropoietic and proliferative defects in MDS.
- ASO depletion of pathogenic *MBD1* transcripts in vitro promotes erythroid differentiation, showing the utility of RNA therapies for MDS.

Myelodysplastic neoplasms (MDS) feature hematopoietic deficits driven, in part, by transcript splicing abnormalities. To date, such disease-driving transcripts have been identified in association with specific splicing factor mutations. However, conserved aberrant splicing-derived transcripts that drive MDS independently of mutational status remain poorly studied, despite representing global therapeutic targets. In this study, we characterize an MDS-associated isoform of methyl-binding-domain 1 (*MBD1*; *MBD1-L*) as a novel member of this class of transcripts. Rather than originating from a mutant splicing factor, the abnormal production of *MBD1-L* is driven by reduced WTAP expression in MDS. Overexpression of *MBD1-L* in healthy human hematopoietic stem and progenitor cells recapitulated archetypal MDS defects, including reduced terminal glycophorin A⁺ erythroid differentiation, suppressed cell cycling, and impaired in vivo reconstitution capacity during increased hematopoietic demand in xenotransplantation assays. An integrated multiomics approach assessing DNA binding of *MBD1* isoforms, and the resulting changes in chromatin accessibility, histone mark deposition, and transcriptional changes, revealed that these defects arise from an isoform-specific switch in *MBD1* binding behavior. The *MBD1-L* isoform refocuses heterochromatin-promoting activity of *MBD1-L* from methylated to unmethylated CpGs, thus enacting broad downregulation of CpG-rich promoters and secondary epigenetic effects mediated through its downstream target, *BCOR*. Remarkably, we also found that directly reversing abnormal *MBD1* splicing across a broad range of primary human MDS samples using nanoparticle-encapsulated antisense RNA oligonucleotides (ASO) enhanced in vitro erythroid differentiation, supporting the therapeutic use of ASO-based therapies for MDS treatment. Thus, our findings demonstrate *MBD1-L* to be a global disease-driving splice variant in MDS and illustrate the potential of RNA-based therapies for the broad treatment of MDS.

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Introduction

Myelodysplastic neoplasms (MDS) are characterized by dysfunctional terminal cell maturation and commitment of hematopoietic stem cells (HSC).¹⁻⁷ Although MDS may evolve into aggressive secondary acute myeloid leukemia, MDS itself also imposes significant disease burden,^{8,9} with non-acute myeloid leukemia mortality being the primary driver of

survival outcomes.¹⁰⁻¹² Erythropoiesis is commonly impaired, which leads to anemia and transfusion dependence. To address this, hypomethylating agents, erythropoietin, or luspatercept are commonly used to either promote healthy hematopoiesis or drive terminal maturation. However, these approaches yield an erythroid response in only ~50% of patients, with even fewer achieving transfusion independence.¹³⁻¹⁶ Furthermore, many patients experience only a

transient erythroid response, leaving limited therapeutic options after treatment failure. This underscores the need to attain a deeper understanding of the molecular underpinnings of MDS with a view towards identifying novel, effective treatment strategies.

Splicing dysregulation is a core feature of MDS, with 60% of cases exhibiting a recurrent splicing factor (SF) mutation in *SF3B1*, *SRSF2*, *U2AF1*, or *ZRSR2*.¹⁷ The resulting alternative splicing (AS) events contribute to hematopoietic dysfunction by altering the function or stability of key target transcripts.¹⁸⁻²¹ Most described AS events occur in a specific SF mutational subtype and are not found in other SF-mutant or wild-type forms of the disease. However, splicing abnormalities may also arise through mutation-independent mechanisms.²² Moreover, signatures of aberrant splicing have general prognostic power across MDS mutational subgroups, including SF-wild-type.²³ This suggests that, in addition to mutation-specific splicing events, MDS may also involve a parallel set of mutation-independent and disease cross-cutting splicing events. Such events may highlight convergent disease-promoting mechanisms and represent promising global therapeutic targets applicable across all patients with MDS. However, few members of this class of alternatively spliced drivers have been identified, and their mechanisms of action remain poorly studied.²⁴

In this study, we identify a sizeable overlap of AS events between SF-mutant and wild-type MDS. We describe the AS of the epigenetic reader methyl-binding-domain 1 (*MBD1*) as one such event, linked to a depletion of the RNA-binding protein WTAP. Through broad rewiring of its chromatin occupancy, misspliced *MBD1* distorts hematopoietic stem and progenitor cell (HSPC) function, which impairs erythropoiesis and responses to hematopoietic demand, thereby mirroring the sustained anemia and cytopenia observed in MDS. Remarkably, reversing excess missplicing of *MBD1* ameliorates MDS erythroid differentiation defects.

Methods

Primary cell culture

Cord blood (CB) and bone marrow (BM) HSPCs were cultured and lentivirally transduced in StemSpan SFEM II (StemCell Technologies, catalog no. 09655) supplemented with 100 ng/mL stem cell factor, 100 ng/mL FMS-like tyrosine kinase 3 ligand, 20 ng/mL interleukin (IL)-6, and 20 ng/mL thrombopoietin (supplemental Methods available on the *Blood* website). CB HSPCs were used for in vitro immunophenotyping, cellular energetics assays, multiomics analyses, and xenotransplantation studies. BM HSPCs were used for in vitro immunophenotyping. MDS HSPCs were cultured in StemSpan SFEM II supplemented with 100 ng/mL stem cell factor, 100 ng/mL FMS-like tyrosine kinase 3 ligand, 20 ng/mL IL-6, 20 ng/mL thrombopoietin, and 10 ng/mL IL-3. For antisense RNA oligonucleotide (ASO) treatment, MDS HSPCs were cultured on BM mesenchymal stromal cells.

In vivo xenotransplantation

NSG (RRID: IMSR_JAX:005557) and NSG-SGM3 (RRID: IMSR_JAX:013062) mice were irradiated with 315 cGy 1 day

before interfemoral transplantation. NSG-W41 mice were not irradiated. NSG, NSG-W41, and NSG-SGM3 mice were used for short- and long-term primary transplantation, whereas NSG-SGM3 mice were used for secondary limiting dilution assays. Transplanted cells were then isolated from the injected femur by crushing and passing through 40µm filters and were analyzed as previously described via flow cytometry.²⁵ Additional detailed methods are available in the supplemental Methods.

Results

A CXXC-3 containing isoform of *MBD1* is globally upregulated in MDS

To examine the potential of non-SF mediated contribution to MDS splicing dysregulation, we analyzed published RNA-sequencing (RNA-seq) data from MDS vs healthy CD34⁺ HSPCs published by Pellagatti et al.²⁶ Interestingly we found that splicing regulators are negatively enriched as a group in MDS regardless of SF mutant status, with *U2AF1* and *SRSF2* transcripts reduced in SF-wild-type MDS (supplemental figure 1A-B).²⁶ Using replicate multivariate analysis of transcript splicing to identify AS events unique to MDS, we revealed >200 AS events displaying conservation across mutational subtypes including SF-wild-type (Figure 1A; supplemental Table 1).^{26,27} To enrich for AS events with functional involvement, we predicted alterations to transcript and protein sequence (Figure 1B; supplemental Table 2).²⁸ We accounted for impacts on translatable mRNA availability, as well as in-frame changes affecting protein domains. Interestingly, we noted an enrichment for gain-of-function splicing changes among AS events, primarily through nonsense-mediated decay rescue or by the inclusion of functional domains (Figure 1C). Focusing on the latter as changes that could be amenable to therapeutic modulation, we identified inclusion of exon 11 (E11) in *MBD1* transcripts as a key event of interest where preference for the E11-containing isoform (*MBD1-L*) over the E11-exclusion form (*MBD1-S*) is found in both SF-mutant and wild-type MDS with minimal change in total *MBD1* abundance (Figure 1D). We validated this enrichment via isoform specific quantitative polymerase chain reaction (qPCR) of primary MDS samples (supplemental Figure 1C).

MBD1 reads methyl-CpGs to establish heterochromatin through recruitment of H3K9 methylase complexes. E11 contains the CXXC-3 domain of *MBD1*, which targets unmethylated CpGs, as opposed to the constitutive MBD domain that binds at methyl-CpGs (Figure 1E).²⁹⁻³¹ CXXC-3 distinguishes *MBD1* from all other MBD family members, which provide functional redundancy for repression of methylated, but not unmethylated, CpGs.³² Although CXXC-3 is dispensable for the repressive activity of *MBD1*, its strong phylogenetic conservation likely suggests a functional significance for this domain (supplemental Figure 1D).^{33,34} *MBD1-L* was enriched in HSCs (CD34⁺CD38⁻CD90⁺CD45RA⁻) and multipotent progenitors (MPPs) (CD34⁺CD38⁻CD90⁻CD45RA⁻), with its expression progressively declining in CD34⁺CD38⁺ progenitors and terminal CD34⁻ cells (Figure 1F). HSCs and MPPs also exhibited increased total *MBD1* mRNA and protein expression (supplemental Figure 1E).³⁵⁻³⁷ Interestingly, *MBD1-L* significantly increases with aging in CD34⁺38⁻ HSPCs, where MDS incidence also increases (Figure 1G).³⁸

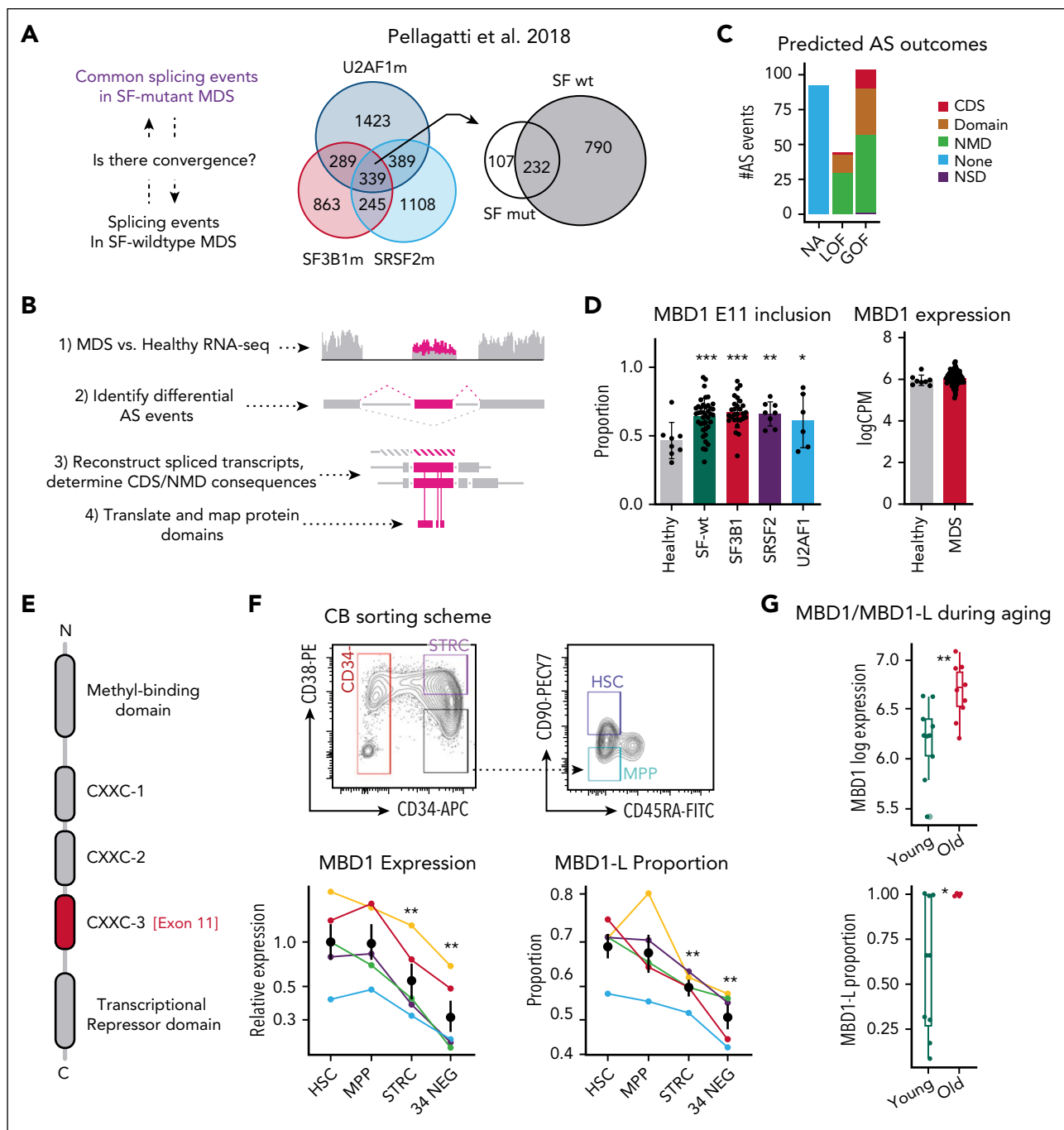


Figure 1. A CXXC-3 containing isoform of MBD1 is globally upregulated in MDS. (A) Replicate multivariate analysis of transcript splicing analysis using the Pellagatti et al²⁶ data set reveals 232 splicing events enriched in CD34⁺ MDS HSPCs that are conserved between splice factor mutant and wild-type subgroups (82 MDS samples; 28 SF3B1 mutant, 8 SRSF2 mutant, 6 U2AF1 mutant, and 40 splice factor wild-type). (B) Workflow for the identification of transcript alterations due to identified differential splicing events. (C) Predicted functional outcomes of MDS-wide AS events, by level of regulation. CDS is defined as the establishment or disruption of a valid CDS. Domain is defined as an alteration of an interpro-annotated domain. Nonsense-mediated decay (NMD) is defined as an alteration in NMD sensitivity. NSD is defined as an alteration in NSD sensitivity. (D) Proportion of MBD1-L in healthy controls or SF-wild-type or mutant MDS subgroups, along with total MBD1 gene expression in healthy and MDS CD34⁺ HSPCs (healthy controls N = 8). (E) Functional domains of MBD1. (F) qPCR of total MBD1 expression or MBD1-L proportion in CB CD34⁺38⁺90⁺45RA⁺ HSCs, CD34⁺38⁺90⁺45RA⁺ MPPs, CD34⁺38⁺ STRCs or terminal CD34⁺ cells (N = 5 samples, paired t test against HSCs). (G) MBD1-L proportion and total MBD1 expression in healthy young and aged CD34⁺38⁺ HSPCs. *P < .05; **P < .01; ***P < .001. CDS, coding sequence; GOF, gain-of-function; LOF, loss-of-function; NA, not applicable; NSD, nonstop decay; STRC, short-term repopulating cell; wt, wild-type.

MBD1-L impairs the response to hematopoietic demand in an isoform-specific manner

We first determined whether MBD1-L imparts MDS-like defects on healthy human HSPCs by lentivirally overexpressing MBD1-L or -S in CB CD34⁺ HSPCs (Figure 2A; supplemental Figure 2A).

We initially included a control for E11 sufficiency in the form of a second variant of MBD1-L with E13 excluded (MBD1-L*). In vitro competitive cultures showed a rapid depletion of MBD1-L-overexpressing cells, compared with a mild depletion for MBD1-S (Figure 2B). Importantly, MBD1-L and MBD1-L*

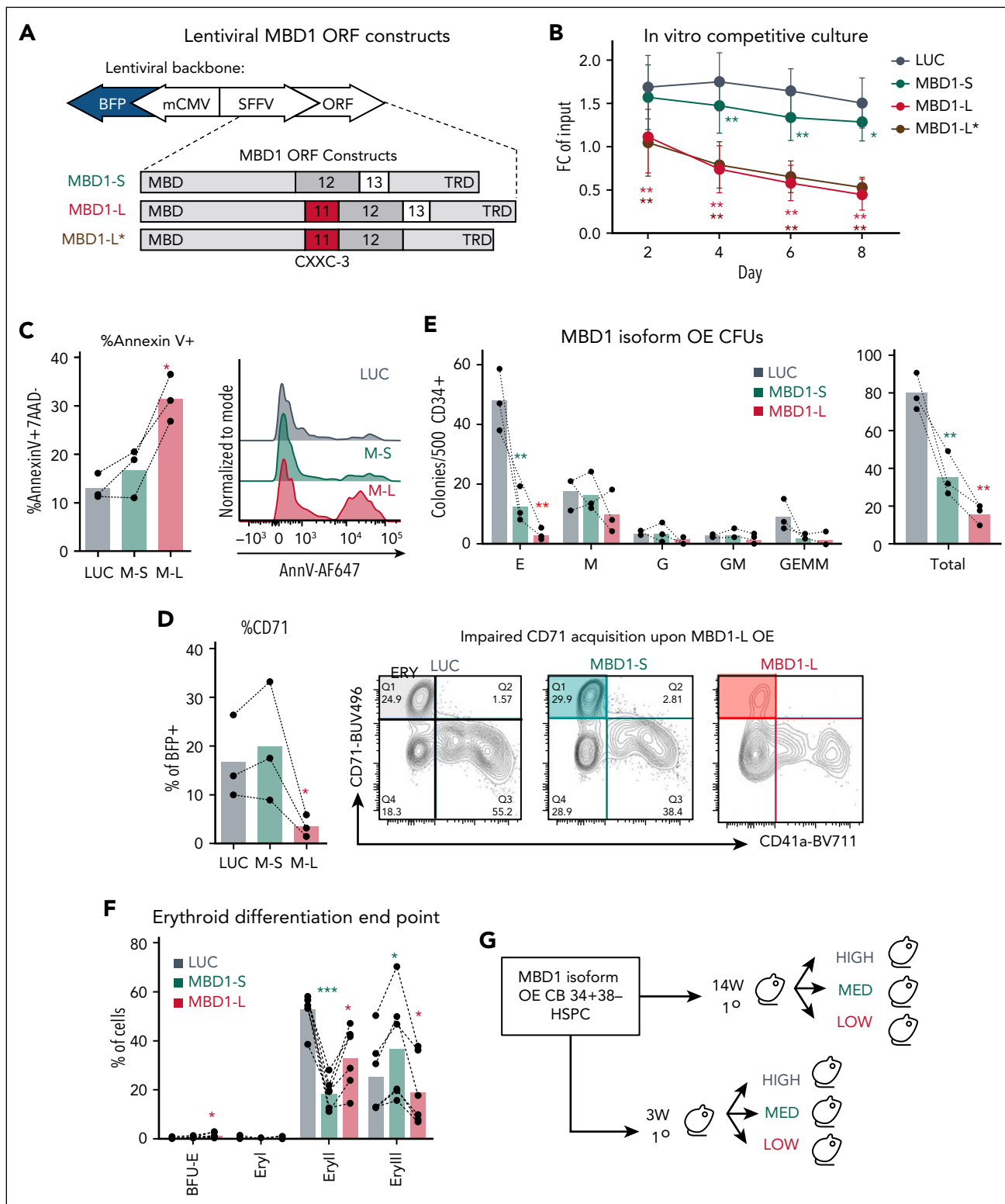


Figure 2. MBD1-L impairs the response to hematopoietic demand in an isoform-specific manner. (A) Schematic of MBD1 isoform lentiviral overexpression constructs. (B) In vitro competitiveness of cultured MBD1 isoform or control overexpressing CB Lin⁺CD34⁺ HSPCs. (C) In vitro annexin V and 7AAD apoptosis analysis in transduced Lin⁺CD34⁺ HSPCs. (D) CD71 expression upon MBD1 isoform overexpression in Lin⁺CD34⁺ HSPCs. (E) Colony output of MBD1 isoform expressing HSPCs. (F) Proportion of erythroid lineage populations in MBD1 isoform overexpressing Lin⁺CD34⁺ HSPCs at the end point of a 2-step erythroid differentiation assay (N = 6 samples). (G) Schematic of 3-week and 14-week competitive transplants of isoform overexpressing Lin⁺CD34⁺38⁻ HSPCs into NSG mice, followed by secondary transplants at limiting dilution. (H) Output/Input competition analysis within the injected femur of 3- and 14-week primary NSG xenografts. (I) CD11b proportion in primary 14-week NSG xenografts. (J) ELDA calculated HSC frequencies in pooled 3- and 14-week primary xenograft BM. (K) Output/Input competition analysis within the injected femur of primary NSG-SGM3 and W41 transplants. (L) GlyA proportion and GlyA⁺/EryL ratio within the injected femur of 3-week primary NSG xenografts. (M) GlyA proportion and GlyA⁺ to EryL ratio within the injected femur of primary W41 xenografts. (N) Summary of hematopoietic impairment upon MBD1-L overexpression in healthy HSPCs. N = 3 samples for in vitro assays

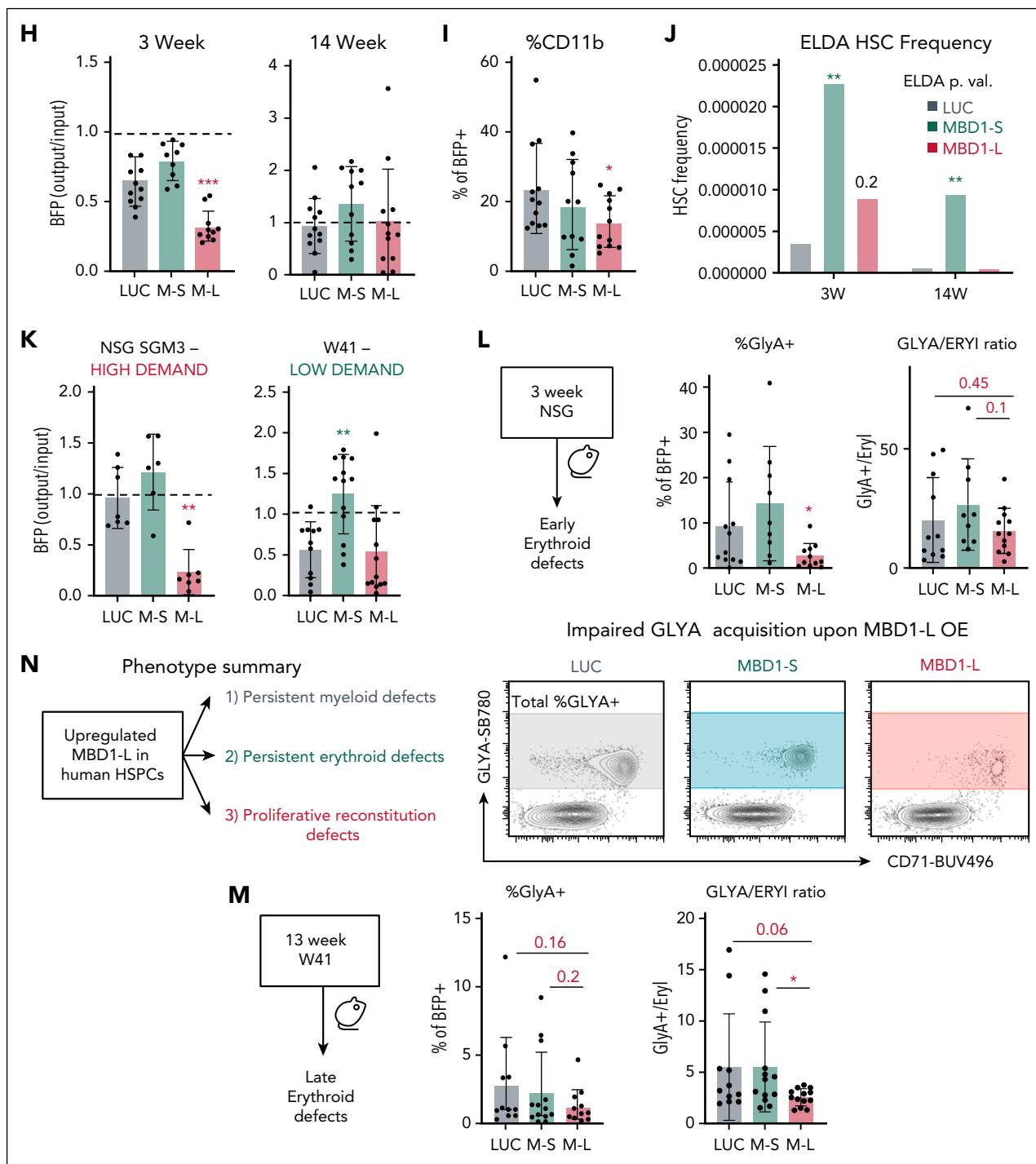


Figure 2 (continued) unless stated otherwise. All transplant data shown is from analysis of the injected femur unless stated otherwise, where each data point represents a mouse. All statistical tests are performed against luciferase unless otherwise indicated. A paired t test was performed for panels B-F, and an unpaired t test was performed for panels H-I and K-M, where equal variances were not assumed. For panel J, the elda R package was used to determine HSC frequencies based on binary engraftment calls per mouse, where differences between conditions were assessed through a χ^2 test. * $P < .05$; ** $P < .01$; *** $P < .001$. BFP, blue fluorescent protein; CFU, colony forming unit; E, erythroid; ELDA, extreme limiting dilution assay; FC, fold change; G, granulocyte; GEMM, granulocyte-erythroid-megakaryocyte-macrophage; GM, granulocyte-macrophage; LUC, luciferase; M, macrophage; M-L, MBD1-L; M-S, MBD1-S; MED, medium; OE, overexpression; ORF, open reading frame; TRD, transcriptional repression domain.

produced identical phenotypes, indicating that CXXC-3 is specifically responsible for the observed defects. In addition, we observed increased annexin V, as well as impaired erythroid differentiation as observed by a decrease in CD71 with no change in myeloid differentiation (Figure 2C-D; supplemental Figure 2B). Colony-forming assays showed decreased burst

forming unit-erythroid (BFU-E) and granulocyte colony-forming unit (CFU-G) colonies upon MBD1-L overexpression, with a magnitude more pronounced than that observed for MBD1-S (Figure 2E). Because differentiation defects were primarily in the erythroid lineage and are prevalent in MDS, we performed erythroid differentiation cultures, measuring erythroblast (EryI)

and reticulocyte (EryIII) output. MBD1-L impaired terminal erythroid cell production by proportion and cell count, whereas MBD1-S increased the proportion of mature EryIII cells but with a mild decrease in cell count, implicating MBD1-L as a driver of dysfunctional erythropoiesis in MDS (Figure 2F; supplemental Figure 2C). To rule out confounding effects from residual endogenous *MBD1* transcripts, we repeated our overexpression experiments with simultaneous depletion of endogenous MBD1 via a cotransduced short hairpin RNA (shRNA), which recapitulated the proliferative and erythroid deficits associated with MBD1-L (supplemental Figure 2D-E). We also observed similar phenotypes when MBD1-L was overexpressed in adult BM CD34⁺ HSPCs (supplemental Figure 2F).

We next assessed MBD1 isoforms in vivo using competitive xenotransplantation of CB Lin[−]CD34⁺CD38[−] HSPCs into NSG mice (Figure 2G). Tracking the proportion of transduced human cells over time revealed that MBD1-L hindered competitiveness during the proliferative initial phase of engraftment at 3 weeks but conferred no disadvantage in the steady state grafts at 14 weeks (Figure 2H; supplemental Figure 2G-I). Nonetheless, we observed reduced CD11b in late-stage MBD1-L grafts, highlighting persistent impairments in myeloid differentiation (Figure 2I). We then transplanted 3- and 14-week grafts into secondary recipients at limiting dilution to measure changes in HSC frequency and observed no difference between control and MBD1-L cells, suggesting that early MBD1-L competitive defects arise from reduced function rather than loss of HSCs (Figure 2J; supplemental Figure 2J-K). Interestingly, we observed a 7- and 11-fold increase in HSCs upon MBD1-S overexpression at 3 and 14 weeks, respectively, supported by increased engraftment and percentage of CD34⁺ levels in secondary recipients (supplemental Figure 2L).

To further explore reconstitution defects of MBD1-L, specifically at proliferative stages of reconstitution, we performed primary transplants into nonirradiated W41 mice, which provide a low-demand environment with no irradiation-induced BM ablation, or irradiated NSG-SGM3 mice, in which transgenic overexpression of human cytokines drives chronic proliferation. The graft-suppressing effect of MBD1-L persisted into the long term in NSG-SGM3 but not in W41 mice, suggesting that excess MBD1-L hinders the proliferative response during conditions of high hematopoietic requirement (Figure 2K).

Finally, we assessed impairments in erythroid differentiation in vivo. As erythroid output occurs in NSG mice at ~3 weeks, we assessed the erythroid graft present at this time and observed that MBD1-L, but not MBD1-S, compromised mature erythroid output (GlyA⁺; Figure 2L). Given that W41 mice model long-term human erythropoiesis in vivo, we assessed terminal erythroid output at a late 13-week time point. By measuring the ratio of terminal GlyA⁺ cells to GlyA[−]CD71⁺ progenitors, we confirmed that MBD1-L-overexpressing HSPCs had reduced capacity to produce terminal erythroid cells in contrast to control and MBD1-S cells, with a trend toward total reduced GlyA (Figure 2M). Altogether, our results suggest that MBD1-L drives myeloid, erythroid, and proliferative defects, possibly contributing to the inability of MDS marrow to increase cell output in response to proliferative signals associated with cytopenia (Figure 2N).

MBD1-L broadly represses promoter CpG islands

Because our results supported a dominant function for CXXC-3, we hypothesized that CXXC-3 recruits the heterochromatin-forming complex of MBD1-L to new genetic loci which are inaccessible to MBD1-S.³⁹ We assessed the nuclear distribution of MBD1 isoforms via immunofluorescence microscopy in CB CD34⁺ HSPCs overexpressing MBD1-S or MBD1-L. MBD1-S localized near the nuclear periphery, which is rich in H3K9me3 heterochromatin, whereas MBD1-L instead localized towards the nuclear interior (Figure 3A). This suggests that MBD1-S predominantly associates with constitutive heterochromatin, whereas MBD1-L acts on transcriptionally active genic regions within the nuclear interior.⁴⁰ To map this in greater detail, we performed CUT&RUN on CB HSPCs overexpressing the MBD1 isoforms (supplemental Table 3). Mapping peaks onto published and unpublished enzymatic methyl-converted HSPC DNA methylation profiles revealed that MBD1-L preferentially binds CpG rich sites, whereas MBD1-S preferentially binds methyl-CpGs, confirming that CXXC-3 actively shapes the DNA-binding footprint of MBD1-L (Figure 3B).⁴¹ In addition, the inclusion of CXXC-3 triggers a striking redistribution of MBD1-L to proximal promoters, likely driven by the presence of promoter-associated CpG islands (Figure 3C; supplemental Figure 3A). MBD1-L bound promoters were enriched in cell cycle-related genes, implicating cell proliferation as a likely control mechanism mediating the hematopoietic defects induced by MBD1-L (supplemental Figure 3B).

We then asked whether the inclusion of CXXC-3 alters MBD1's associated protein complex by performing Rapid Immunoprecipitation Mass Spectrometry of Endogenous Protein (RIME) profiling of chromatin-associated complexes in MDS-L cells.⁴² Analysis of the MBD1-L or MBD1-S interactomes revealed that both isoforms associated with largely the same set of nuclear proteins (Figure 3D; supplemental Figure 3C; Table 4). However, we did observe quantitative differences in the level of association. MBD1-L exhibited increased association with its canonical SETDB1 H3K9 methylase partner complex but exhibited reduced association with components of constitutive heterochromatin, including the HP1 proteins CBX5 and CBX3, in agreement with the altered physical distribution of MBD1-L away from heterochromatic regions (Figure 3E).⁴³⁻⁴⁷

To explore whether MBD1-L closes bound hypomethylated promoters, we used assay for transposase-accessible chromatin (ATAC)-sequencing and observed that MBD1-L indeed produces global repression of ATAC signal at promoter regions, whereas MBD1-S does not (supplemental Table 5). As expected, ATAC signal at intronic enhancers, which lack extensive CpG islands, showed no differences between MBD1-S and MBD1-L (Figure 3F). Furthermore, the level of regional ATAC repression by MBD1-L correlated with its CpG density, indicating that the most CpG-rich promoters exhibit greater sensitivity to repression by MBD1-L (Figure 3G). Importantly, we identified that MDS HSPCs exhibit a negative enrichment of genes with CpG-rich promoters compared with healthy controls, suggesting that the observed pattern of MBD1-L repression is active in MDS (Figure 3H).²⁶ Finally, we validated that the global closure of promoter regions by MBD1-L was associated with reduced transcription by measuring

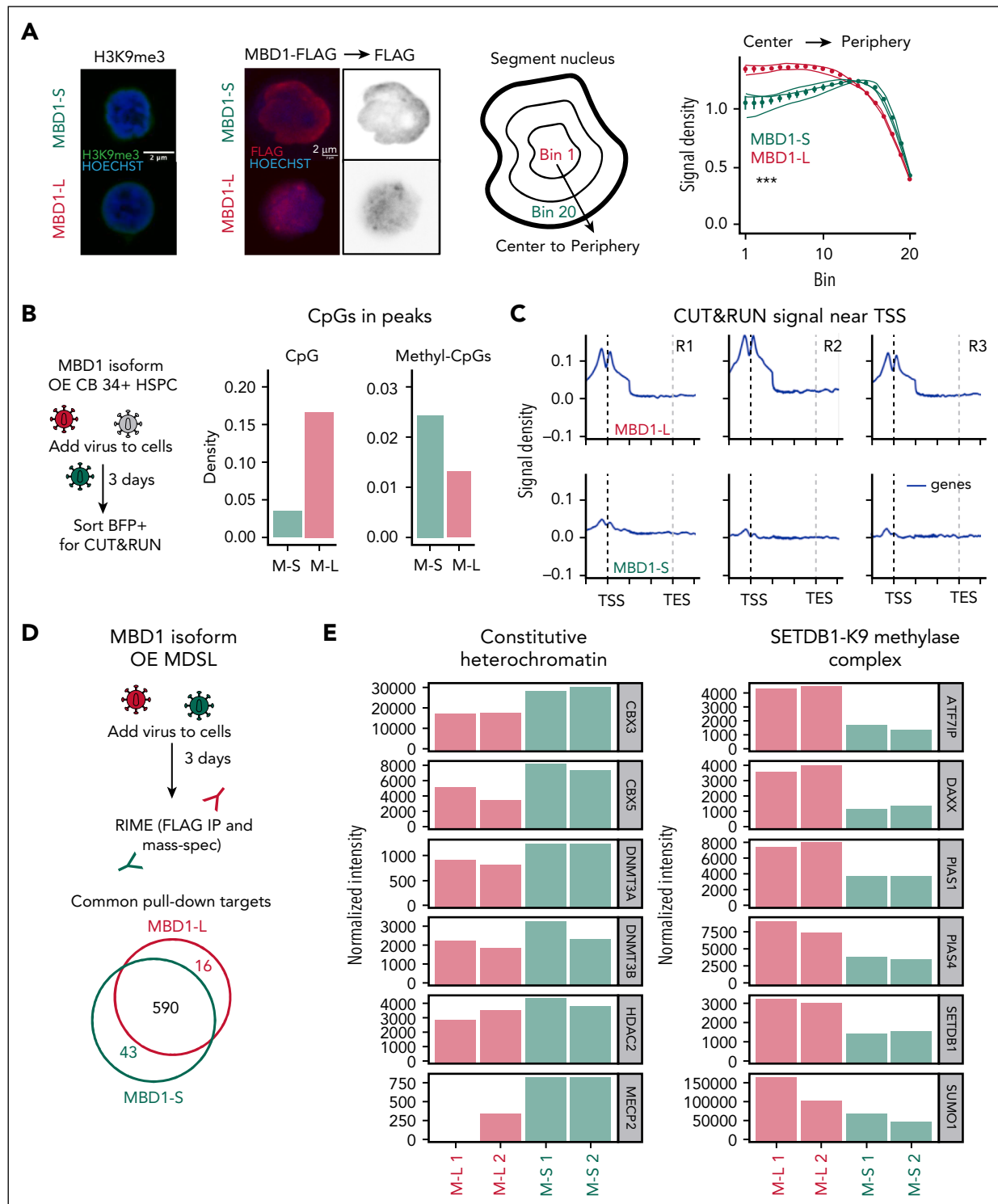


Figure 3. MBD1-L broadly represses promoter CpG islands. (A) H3K9me3 and MBD1 (via FLAG tag) staining in Lin⁺CD34⁺ HSPCs transduced with MBD1-L and MBD1-S lentivectors. FLAG staining was used to calculate the distribution of MBD1 isoforms within the nuclear area, measured 72 hours after transduction (N = 30 analyzed cells per sample, 3 biological samples, statistical test was a multivariate analysis of variance using biological replicate as a covariate). (B) Estimated CpG and methylated CpG density within CUT&RUN peaks. (C) Peak distribution across gene regions in MBD1-L and MBD1-S CUT&RUN. (D) Schematic of RIME analysis of the MBD1-L and MBD1-S repressive complex in MDSL, with 590 targets overlapping between immunoprecipitated interacting proteins between the 2 isoforms. (E) Comparison of signal intensities of detected RIME-binding partners in MDSL. (F) Differential ATAC accessibility at intronic enhancers and promoters for MBD1 isoforms vs luciferase. The promoter group has been partitioned into CpG-rich and CpG-poor subsets based on a 5% CpG cut-off. (G) Correlation between ATAC peak repression by MBD1 isoforms and CpG density within the peak. (H) GSEA enrichment of the top 250 CpG-dense promoters in the Pellagatti et al²⁶ MDS Lin⁺CD34⁺ MDS HSPC RNA-seq. (I) H3K9me3 CUT&RUN signal proximal to the TSS at MBD1-L bound and repressed promoters compared with luciferase. All multiomics experiments were performed with 3 independent CB samples. ***P < .001. BFP, blue fluorescent protein; M-L, MBD1-L; M-S, MBD1-S; mass-spec, mass-spectrometry; R1, replicate 1; R2, replicate 2; R3, replicate 3; TES, transcriptional end site; TSS, transcriptional start site.

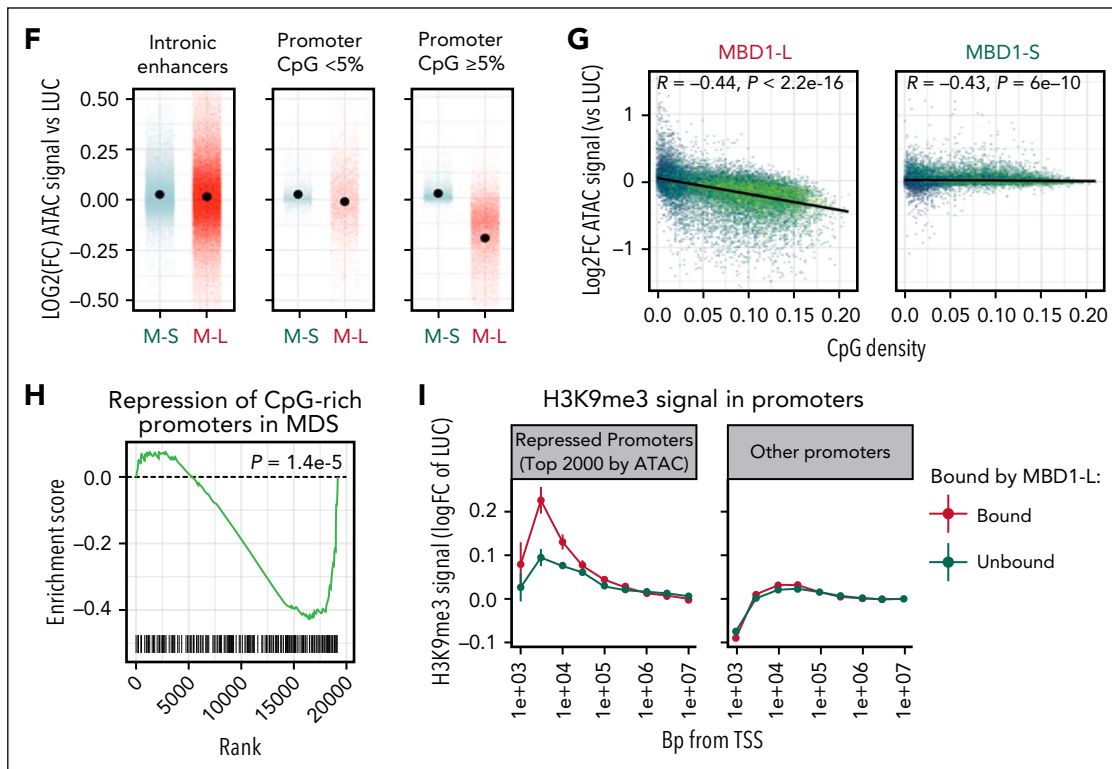


Figure 3 (continued)

ethynyl uridine integration, where MBD1-L inhibited the S-phase associated increase in transcription (supplemental Figure 3D).

Given that MBD1-L binds and represses promoter CpG islands, it stands to reason that these promoters would contain higher levels of repressive H3K9me3 marks. We found that overexpression of MBD1-L in HSPCs did not trigger an overall shift of H3K9me3 into the nuclear interior (supplemental Figure 3E) and reasoned that H3K9me3 deposition induced by MBD1-L may be more subtle. We therefore profiled H3K9me3 and the activating H3K4me3 histone marks at promoters using CUT&RUN on MBD1-S- or MBD1-L-overexpressing CD34⁺ HSPCs. Promoters containing MBD1-L binding peaks, representing direct loci of inhibition, indeed accumulated H3K9me3 marks within 1 to 10 kb of the transcription start site upon MBD1-L overexpression (Figure 3I; supplemental Table 6). Conversely, MBD1-L binding exhibited no correspondence with H3K4me3 status, confirming that closure of promoters by MBD1-L depends on the installation of inhibitory H3K9me3 marks (supplemental Figure 3F).

MBD1-L silences promoter networks to inhibit cell cycling and erythroid differentiation

To investigate the downstream transcriptomic impacts of MBD1-L activity, we performed bulk RNA-seq. Sets of differentially expressed genes were highly distinct between MBD1 isoforms (Figure 4A; supplemental Table 7). In addition, repression of cell cycle-associated gene sets represented the major enrichment observed for MBD1-L, in accordance with similar enrichment observed in its bound promoters (Figure 4B; supplemental Table 8). Indeed, ethynyl deoxyuridine cell cycle

assays revealed that MBD1-L impaired the transition from G1 to S phase (Figure 4C). Importantly, this aligns with the observed downregulation of cell cycle pathways observed in MDS HSPCs (Figure 4D).²⁶ Projection of MBD1-L-repressed targets onto hematopoietic subpopulations revealed a downregulation of genes associated with mature erythroid populations, consistent with our observed erythroid defects (supplemental Figure 4A).⁴⁸ However, we did not observe differential expression of key erythroid regulators, suggesting that MBD1-L may regulate erythroid differentiation at the HSPC level via alternative means (supplemental Figure 4B).

To identify MBD1-L targets representing novel candidate effectors of our erythroid phenotype, we integrated our multi-omics data sets (Figure 4E), filtering for genes exhibiting MBD1-L-unique binding, ATAC-seq repression, and RNA-seq downregulation at a threshold of *P* value of <0.05. Among this set of direct targets was BCL6 corepressor (BCOR; supplemental Figure 4C). BCOR participates in epigenetic silencing as part of the BCL6 repressive complex and Polycomb Repressive Complex 1.1, and its function is impaired in MDS by both recurrent mutations and expression downregulation (supplemental Figure 4D).⁴⁹⁻⁵¹ The BCOR promoter is notable because it is encompassed by a 15.8 kb CpG island, the longest among protein-coding genes, making it particularly sensitive to regulation by MBD1-L (supplemental Figure 4E). This island features a methylated center flanked by hypomethylated regions corresponding to ATAC peaks. MBD1-L binds these flanking regions and suppresses the ATAC signal, accompanied by an increase in H3K9me3 (Figure 4F-G). Furthermore, MBD1-L derepressed BCOR-controlled genes (Figure 4H). Several of the most upregulated genes upon

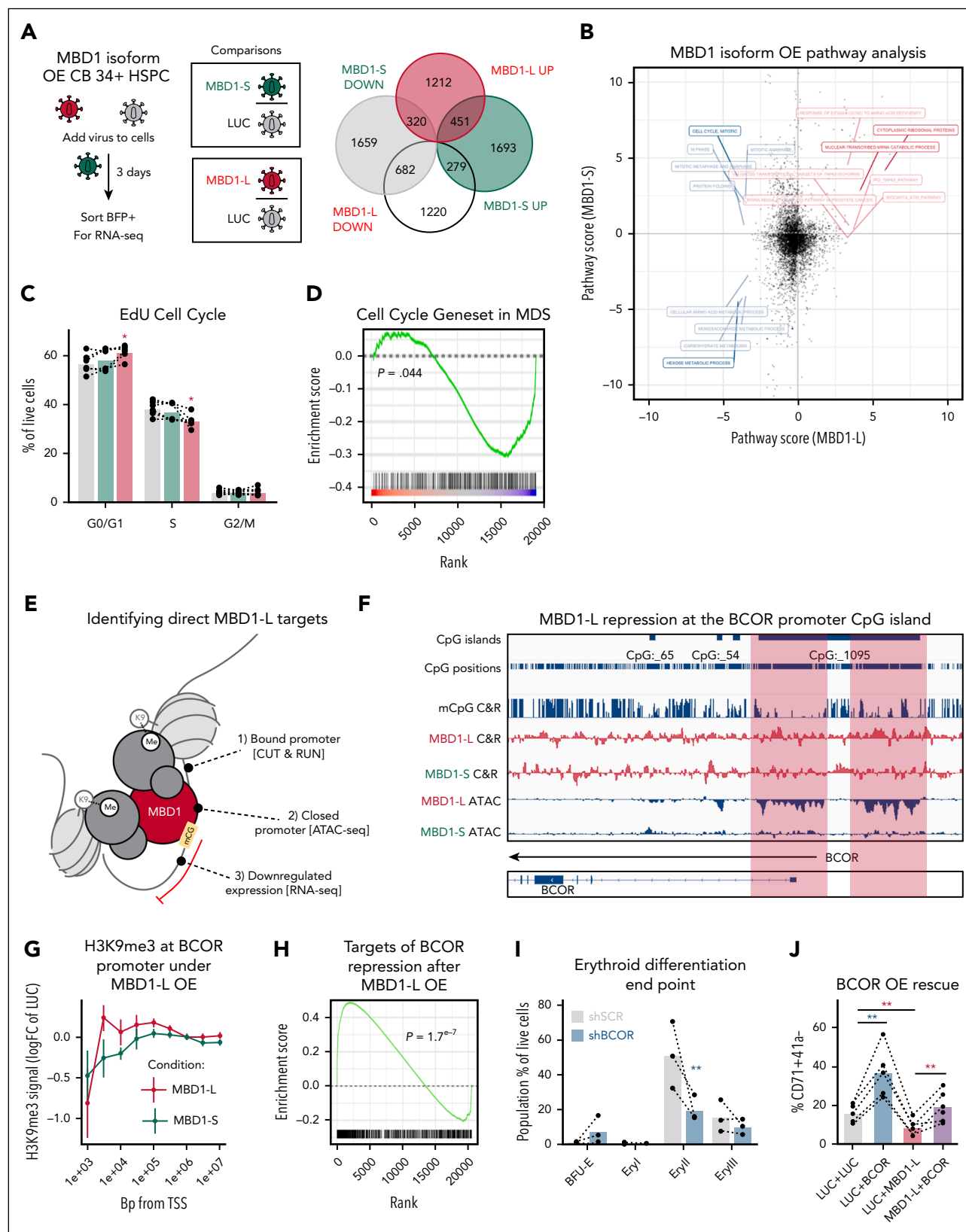


Figure 4. MBD1-L silences promoter networks to inhibit cycling and erythroid differentiation. (A) Overlap in differentially expressed genes between luciferase and either MBD1-S and MBD1-L transduced Lin[−]CD34⁺ HSPCs ($P < .05$ vs luciferase). (B) GSEA enrichments in MBD1-L vs MBD1-S overexpression, labeled with gene sets enriched in MBD1-L overexpression (opaque: $P < .05$; faded: $P < .1$). (C) Intracellular flow cytometry of MBD1 isoform overexpressing Lin[−]CD34⁺ HSPCs pulsed with ethynyl deoxyuridine for 2 hours, measured 72 hours after transduction ($N = 6$ samples, paired t test). (D) Cell cycle gene set enrichment in MDS CD34⁺ HSPCs (Pellagati et al²⁶). (E) Schematic of pipeline used to select downstream direct targets of MBD1-L. Promoters bound by MBD1-L that were found to be less accessible, and their gene product was downregulated at the RNA level, were considered direct targets repressed by MBD1-L. (F) Tracks displaying MBD1-L and MBD1-S CUT&RUN and ATAC-seq signal in the

MBD1-L overexpression are BCOR targets, including *GDF15*, a transforming growth factor- β ligand associated with ineffective erythropoiesis in MDS (supplemental Figure 4F).⁵²⁻⁵⁴ This supports a second level of indirect epigenetic regulation by MBD1-L, whereby derepression of BCOR targets drives up-regulation of a subset of genes in parallel with the broad promoter repression directly exerted by MBD1-L.

We next aimed to determine whether BCOR loss orchestrates MBD1-L erythroid defects. Erythroid differentiation culture revealed lowered EryII output in response to BCOR knockdown in CD34⁺ HSPCs (Figure 4I; supplemental Figure 4G). We then performed rescue experiments to determine whether exogenous re-expression of BCOR was capable of reversing MBD1-L defects. Indeed, restoration of BCOR levels in MBD1-L-overexpressing HSPCs by cotransduction of a BCOR overexpression lentivirus completely rescued the MBD1-L-induced erythroid defect but did not yield any improvement in the proliferative defect (Figure 4J; supplemental Figure 4H-I). This suggests that the proliferative and erythropoietic phenotypes are independent and arise from parallel but distinct mechanisms, with the loss of proliferation resulting from the repression of cell cycle-related promoters by MBD1-L, and the erythroid deficit attributed to secondary epigenetic repercussions stemming from BCOR insufficiency.

The m6A writer WTAP regulates the splicing of MBD1-L

Because MBD1-L splicing is not associated with specific SF mutations, we hypothesized that RNA regulators globally altered across MDS might be responsible for pathological MBD1 splicing. We therefore scored for differentially expressed RNA-binding proteins (RBPs) from MDS CD34⁺ transcriptomes, whose expression correlated or anticorrelated with the inclusion of E11.²⁶ Intriguingly, in the group of RBPs deficient in MDS, WTAP, a component of the N6-methyladenosine (m6A) writer complex, exhibited the strongest negative correlation with E11 inclusion (Figure 5A; supplemental Figure 5A). We also found that WTAP was downregulated in CD34⁺ MDS HSPCs, consistent with increased MBD1-L (Figure 5B). Supporting the role of WTAP in regulating MBD1-L splicing, we found that its knockdown increased the MBD1-L proportion in MDS-L cells. Likewise, knocking down METTL3 and METTL14, the 2 other core members of the m6A writer complex, also increased MBD1-L splicing, suggesting that although MBD1-L splicing in MDS is primarily associated with WTAP depletion, any loss in m6A writer function also biases toward the MBD1-L isoform (Figure 5C; supplemental Figure 5B). Furthermore, inducing a WTAP-depleted state in normal CB HSPCs via knockdown recapitulated the proliferative and erythropoietic defects induced by MBD1-L overexpression (Figure 5D-E). These

results overall support reduced WTAP expression as a key mechanism enforcing inclusion of E11. Interestingly, we also identified a large-scale correlation between the splicing signature induced by knock out of m6A writers WTAP and METTL3 and the MDS splicing signature, suggesting that m6A modifications may be implicated more broadly in the perturbed splicing state of MDS (supplemental Figure 5C).⁵⁵

The correspondence between loss of m6A writer activity and increased E11 inclusion suggests that m6A modification promotes E11 skipping. To identify whether disruption of DRACH motifs can alter MBD1 splicing, we referenced published CB CD34⁺ HSPC meRIP data and calculated meRIP peak enrichment at DRACH motifs. This analysis identified 3 candidate m6A-marked motifs within the E11 downstream intron (sites 136, 137, and 138) (Figure 5F).^{56,57} To test whether m6A disruption within these sites could increase E11 inclusion, we created splicing minigene reporters spanning E10 to E12, with the target m6A sites inactivated by an A>T point mutation. We transduced MDS-L cells with these reporters and performed reverse transcription PCR using reporter-specific primers to assess E11 inclusion levels (Figure 5G). Of the 3 sites tested, the Δ 136 mutation significantly increased the proportion of spliced reporter transcripts containing E11 (Figure 5H). Using single-base elongation- and ligation-based qPCR amplification (SELECT), which detects m6A at single-nucleotide resolution through methylation-sensitive primer extension and ligation, we confirmed that the 136 DRACH motif was indeed methylated in endogenous MBD1 pre-mRNA in MDS-L (Figure 5I).⁵⁸ m6A at this site additionally decreased upon WTAP knockdown. This overall suggests that loss of m6A at this location because of WTAP depletion may drive preferential MBD1-L splicing across MDS.

The WTAP complex has also been shown to preferentially mediate the exclusion of GC-rich exons flanked by introns with high GC and G-quadruplex (G4) content, a behavior attributed in part to WTAP's interaction with G4 proteins.⁵⁹ We thus hypothesized that intronic G4s may contribute to E11 inclusion alongside m6A. Using rG4 detector and G4-screener, we observed a high likelihood of predicted G4 formation in the E11 downstream intron, with identified m6A 136 to 138 sites located within a high quadruplex-forming region (supplemental Figure 6A). Notably, we found that this pattern also generalized to exons preferentially included in MDS or WTAP knockout HSPCs (supplemental Figure 6B). To functionally assess the role of G4 structures in MBD1 E11 inclusion, we used CRISPR/Cas9-mediated homology-directed repair to knock in a modified sequence into the E11 downstream intron, designed to weaken G4 formation within the region predicted to have the highest G4 probability. Remarkably, weakening the G4 motif led to a decrease in MBD1-L isoform levels in K562 cells, supporting the

Figure 4 (continued) BCOR promoter CpG island. CUT&RUN data is displayed as averaged peaks with immunoglobulin G subtraction from the FLAG IP. ATAC-seq peaks are displayed as averaged logFE compared with luciferase controls. CpG methylation status is also highlighted using an enzymatic methyl-converted CUT&RUN data set. (G) H3K9me3 CUT&RUN signal proximal to the BCOR transcription start site in MBD1-L and MBD1-S overexpressing Lin⁻CD34⁺ HSPCs. (H) Gene set enrichment analysis (GSEA) of BCOR-repressed targets in MBD1-L overexpression RNA-seq (bound targets obtained from Schaefer et al.⁴⁹). (I) Proportion of erythroid lineage populations in BCOR knockdown Lin⁻CD34⁺ HSPCs at the end point of a 2-step erythroid differentiation assay. (J) Rescue of the MBD1-L erythroid defect via BCOR coexpression. Percentage CD71⁺41a⁻ proportion in luciferase transduced Lin⁻CD34⁺ HSPCs that are either cotransduced with luciferase or a BCOR overexpression construct to establish baseline BCOR effects, or MBD1-L to establish the baseline erythroid defect, as well as Lin⁻CD34⁺ HSPCs transduced with MBD1-L and BCOR to measure restoration of erythroid differentiation (N = 6 samples). All multiomics experiments were performed with 3 independent CB samples. All in vitro assays are performed with 3 samples unless otherwise stated. Paired t tests were performed for panels C, I-J. *P < .05; **P < .01. BFP, blue fluorescent protein; C&R, CUT&RUN; EdU, ethynyl deoxyuridine; LUC, luciferase; OE, overexpression.

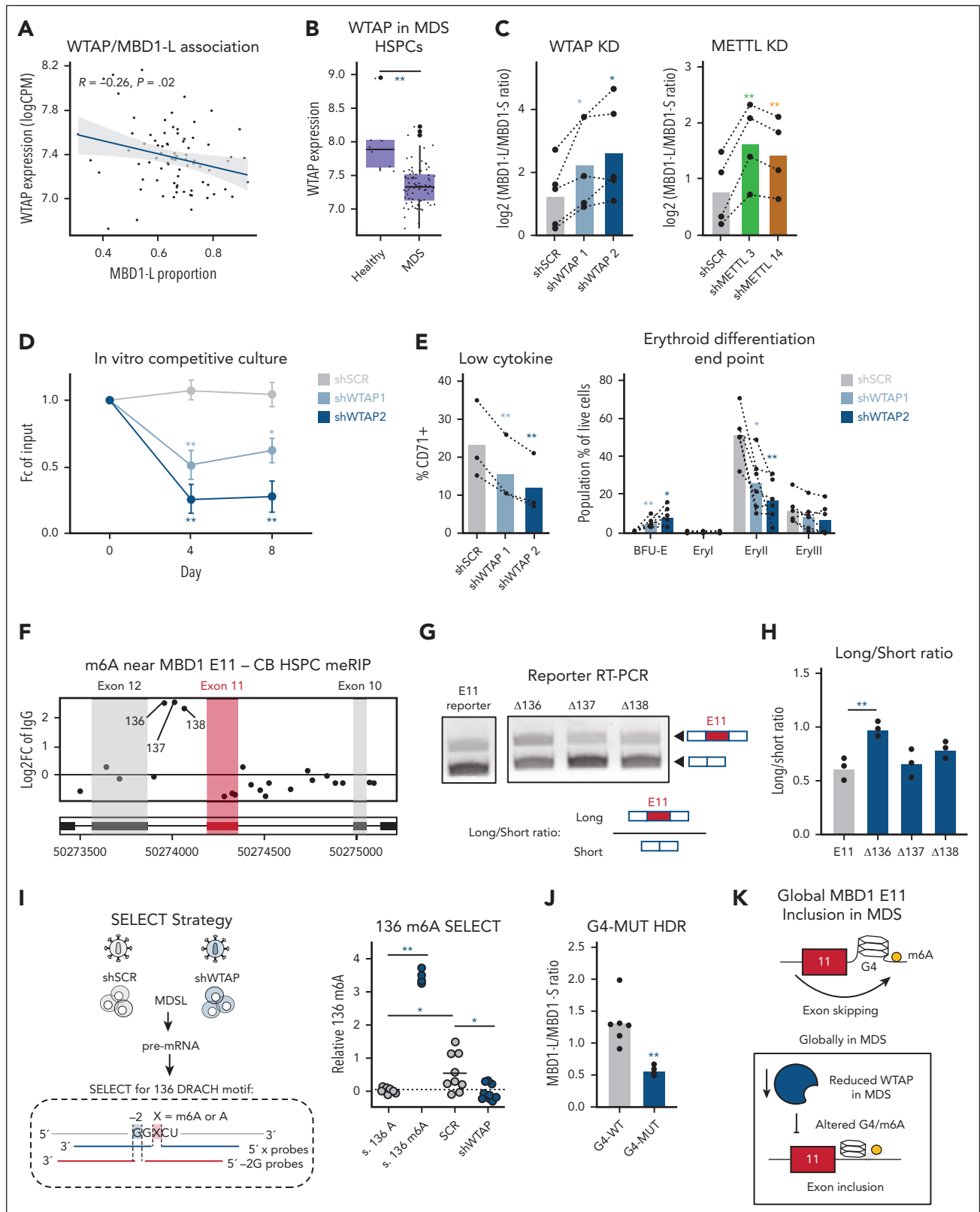


Figure 5. The m6A writer WTAP regulates the splicing of MBD1-L. (A) Correlation between MBD1-L inclusion and WTAP expression in MDS CD34⁺ HSPCs (Pellagatti et al.²⁶). (B) Expression of WTAP in healthy vs MDS CD34⁺ HSPCs (Pellagatti et al.²⁶). (C) qPCR quantification of MBD1-L to MBD1-S ratio 6 days after transduction with shRNAs against WTAP (N = 5 biological replicates), or METTL3 and METTL14 (N = 4 biological replicates) in MDSL. (D) In vitro competitiveness of cultured Lin[−]CD34⁺ HSPCs transduced with shRNAs against WTAP. (E) CD71 proportion in regular cytokine culture upon WTAP knockdown (N = 3 samples) and proportion of erythroid lineage populations in WTAP knockdown Lin[−]CD34⁺ HSPCs at the end point of a 2-step erythroid differentiation assay (N = 6 samples). (F) MerIP peak enrichment at DRACH motifs along the MBD1 transcript in CB CD34⁺ HSPCs (Li et al.⁵⁶). (G) Representative reverse transcription (or transcriptase) PCR of mini-splicing reporter derived MBD1 transcripts in MDSL. (H) E11 mini-splicing reporter quantification of the ratio between the E11 inclusion and exclusion forms. (I) MDS-L were transduced with control or WTAP shRNA, and

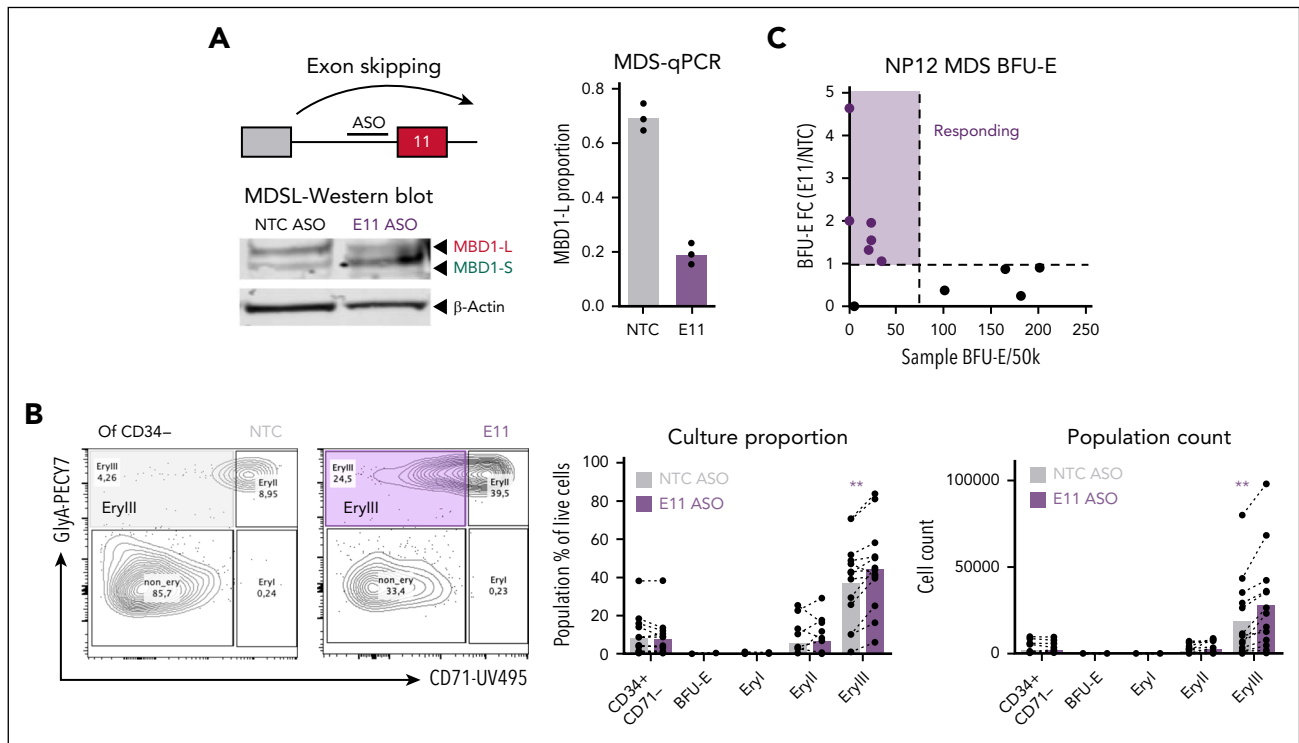


Figure 6. Depletion of MBD1 restores differentiation in MDS cells. (A) Schematic of E11 targeting ASOs impeding E11 inclusion. MBD1-L proportion measured via qPCR after a pulse of control (NTC) or E11 ASO in MDS CD34⁺ HSPCs using ASO packaged in LNPs with a N/P ratio of 12:1 (N = 3 technical replicates). Additional ASO validation was performed using a western blot of MBD1-L and MBD1-S in MDSL cells treated with control and E11 ASOs. (B) Restoration of terminal EryIII proportion and absolute count in primary MDS CD34⁺ HSPCs treated with E11 or control ASOs at the day 18 end point of a 2-step erythroid differentiation assay with MDS HSPCs cultured on a primary human BM mesenchymal stromal cell layer (N = 13 samples, paired t test). (C) BFU-E colonies of primary MDS cells pulsed with E11 or control ASO. Fold change in BFU-E colonies compared with the NTC control are plotted against total BFU-Es in NTC samples. Samples with a BFU-E fold change >1 are highlighted as responding samples (N = 11 samples). All functional assays are performed with 3 samples unless otherwise stated. **P < .01.

idea that G4 structures are required for E11 inclusion and that WTAP could modulate this process (Figure 5J; supplemental Figure 6C). Thus, we identified a confluence of m6A and G4 splicing regulation to mediate MBD1 E11 inclusion (Figure 5K).

Depletion of MBD1 restores differentiation in MDS cells

Given that excess MBD1-L imposes significant hematopoietic impairment, we asked whether reducing MBD1-L can rescue the defective hematopoietic output and erythroid differentiation observed in MDS cells. Because MBD1 disruption in MDS affects solely splicing and not total MBD1 abundance, we elected to directly modulate E11 splicing to simulate restoration of a healthy splicing state while maintaining normal MBD1 levels by using an ASO which sterically inhibits SF recognition of the 3' splice acceptor site of E11 and prevents E11 inclusion (Figure 6A).⁶⁰ To optimize uptake and delivery, we packaged E11-targeting ASOs into lipid nanoparticles (LNPs) using the current benchmark formulation for LNP-based small RNA

delivery.⁶¹ Testing a variety of nanoparticle-payload proportions calibrated by amine/phosphate (N/P) ratio to mediate delivery of RNA cargo into CD34⁺ HSPCs, we found the N/P ratio of 12:1 produced efficacious RNA delivery, through detection of intracellular fluorescein amidite-conjugated control RNAs, as well as MBD1 splicing modulation in MDS cells upon E11 targeting ASO treatment (Figure 6A; supplemental Figure 7A).

To assess the therapeutic potential of the validated E11 ASO, we isolated CD34⁺ MDS HSPCs from 13 patients who were treatment-naïve and cultured cells on human BM mesenchymal stromal cells, then treated with ASO on days 0, 6, and 12 of a 2-step erythroid culture. Remarkably, we found that the E11 ASO increased the proportion and number of late erythroid cells at the culture end point across the majority of tested samples compared with a nontargeting (NTC) ASO (Figure 6B, supplemental Figure 7B-C). We likewise assessed erythroid progenitor function on MDS cells from 12 patients after a

Figure 5 (continued) SELECT was performed on MBD1 pre-mRNA. Probes either targeted the 136 A base, or at an unrelated G base 2 nucleotides upstream. Synthetic RNA oligos with no m6A modification (s.136A) or with m6A modification (s.136 m6A) were used to validate the SELECT method and s.136A was used to establish a baseline for no m6A modification, against which control (SCR) or WTAP shRNA transduced MDS-L were compared to identify the presence of an m6A mark (N = 9 reactions for s.136A and s.136 m6A, and N = 9 samples for SCR and shWTAP MDS-L). (J) qPCR quantification of MBD1-L to MBD1-S ratio upon CRISPR/Cas9 mediated HDR in k562 cells weakening G4 structures (G4-MUT) within the E11 downstream intron compared with wild-type controls (G4-WT) (N = 6, G4-WT samples and N = 5, G4-MUT HDR samples). (K) Proposed mechanism of global increase in MBD1 E11 inclusion in MDS. All functional assays are performed with 3 samples unless otherwise stated. Paired t tests were performed for panels C-E, and an unpaired t test for panels H-J where equal variances were not assumed. *P < .05; **P < .01. HDR, homology directed repair; KD, knockdown; MUT, mutant; SCR, shScrambled; WT, wild-type.

24-hour ASO treatment. This brief ASO treatment yielded an increased erythroid colony output specifically for samples that displayed low basal erythroid output, suggesting MBD1-L depletion may be most beneficial for those cases exhibiting severe erythropoietic impairment (Figure 6C). Separately, we observed a similar increased erythroid colony output in MDS samples with impaired basal erythropoiesis using a less efficacious ASO formulation tested in additional 10 primary samples, further supporting the therapeutic potential of E11 ASO (supplemental Figure 7D).

Discussion

MDS therapies are limited and often fall short of achieving durable disease control. Recurrent AS disruption in MDS offers an underexplored set of therapeutic targets in the form of disease-driving transcript products. Although the majority of described functional instances of pathogenic AS result from SF mutations, we pursued the novel approach of identifying common underlying driver AS events shared across SF-mutant and SF-wild-type MDS, under the hypothesis that these AS events may represent global MDS therapeutic targets. In doing so, we uncovered MBD1-L as a key pathogenic isoform. The repressive action of MBD1-L at promoters inhibits cell cycle processes to reduce hematopoietic output, whereas the specific repression of the BCOR promoter impairs erythroid differentiation. As cycling rate regulates erythroid commitment and maturation, impaired cell cycling may additionally contribute to the MBD1-L-induced erythroid deficit.⁶² Dysregulated cell cycling and increased apoptosis are hallmark features of ineffective erythropoiesis in MDS, and the dampening of responsiveness to proliferative signaling by MBD1-L-overproduction may underlie the marrow's deficient response to anemic stress.⁶³⁻⁶⁵

Interestingly, similar deficits in erythroid and cell-cycle processes are observed in murine models of *Tet2* haploinsufficiency. Because *TET2* depletion represses promoters by impairing the erasure of DNA methylation, these phenotypes may represent conserved sequelae of aberrant CpG island repression in MDS.⁶⁶ In addition, we observed an isoform switch from MBD1-S to -L in HSPCs with aging, where MBD1-S was responsible for reinforcement of constitutive heterochromatin, and this enhances reconstitution and promotes quantitative HSC expansion. Increased MBD1-L with aging may degrade the capacity to maintain organized heterochromatin, driving functional decline and possibly increasing the risk of leukemogenesis. Indeed, disruptions to the abundance and spatial distribution of H3K9me3 are observed in HSPCs upon aging and can contribute to aging phenotypes.⁶⁷⁻⁶⁹ To this point, we found that MBD1-L migrates to central genic regions where H3K9me3 marks spread to active promoters with low CpG methylation. Our study thus showcases preferential production of MBD1-L with age and dominant capacity to alter epigenetic landscapes in ways that mirror aged and/or preleukemic HSPCs.

We found WTAP to have the strongest negative correlation with E11 inclusion amongst MDS-depleted RBPs, suggesting that loss of the m6A writer complex contributes to MBD1-L upregulation. Indeed, inactivation of a DRACH motif within the E11 downstream intron enhanced E11 inclusion within a minigene splicing reporter system, along with a remarkable

correspondence between MDS and WTAP or METTL3 knockout splicing signatures. In addition, we found MBD1 E11, as well as exons included broadly in MDS, to be flanked by introns enriched in predicted G4 structures where the WTAP complex has been shown to associate with G4-binding proteins and exclude similarly structured exons.⁵⁹ Factors governing m6A and G4 control of splicing remain poorly understood, however as our identified DRACH motifs fall within predicted G4 structures and an interplay between m6A and G4 exists in splicing regulation, E11 splicing is likely a confluence of multiple mechanisms hinging on the m6A writer complex and G4 structures.⁷⁰⁻⁷² Characterization of the extent of this could have far reaching impacts on our understanding of splicing disruption in MDS and on its treatment, and as such it will be imperative to functionally evaluate G4s at sites of exon inclusion in MDS, intronic m6A status, and how m6A is regulated in these regions via WTAP.

ASOs represent an attractive mode of mRNA-targeted therapy and have been successfully deployed in the mitigation of various diseases.⁷³ In a first-time application of ASOs in MDS, delivered through an Food and Drug Administration-approved LNP formulation, we show that re-equilibration of the MBD1-L to MBD1-S ratio in patient samples results in a therapeutic increase in erythroid output. A limited 24-hour exposure of MDS HSPCs to the ASO was also sufficient to ameliorate impaired erythropoiesis at the level of progenitors in patients with particularly impaired erythropoiesis, demonstrating the powerful effects of MBD1-L targeting ASOs and sets the stage for further exploring this approach in MDS therapeutics independently or in synergy with interventions targeting terminal red cell maturation, such as luspatercept.^{4,12,15,65} Given the variability in responses across the MDS samples that were tested, it will be important to further assess our ASO in a larger primary sample cohort to further evaluate the therapeutic response in particular within MDS cases refractory to erythropoiesis-stimulating agents, as well as adaptation into the in vivo setting. Because MBD1-L acts in a methylation-independent manner, we expect that any deleterious effects resulting from its epigenetic influence would be intractable by hypomethylating agents (HMA) currently used as first-line MDS therapies. A corollary of this is that increased MBD1-L splicing may be a contributory mechanism to HMA-refractory anemia in MDS and could potentially be an indicator for poor HMA response, whose predisposing factors are not currently well understood.

Our study highlights an extensive landscape of aberrant AS across MDS, including gain-of-function events which represent untapped therapeutic targets, with MBD1-L representing one such key AS event. ASO-mediated correction of MBD1 splicing increases erythroid output in samples from patients with MDS, highlighting a novel MDS therapy well positioned for further development.

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Authorship

Contribution: H.T.T.C. and P.J. designed and performed experiments, analyzed data, interpreted results, and wrote the manuscript; S.C., S.J., J.X., E.T., S.A.A., M.X., and Z.B. designed and performed experiments, as well as analyzed data and interpreted results; D.K. and A.D., under the supervision of K.S.B. and A.N.H., performed and analyzed the MBD1 isoform RIME; Y.M., under the supervision of G.Z., generated and performed quality control on the MBD1 ASO LNPs; D.G.-P., K.C., A.M., O.B., R.S., A.L., N.K., D.C., S.U., T.K., and S.C., under the supervision of H.T., sourced and banked primary MDS samples; M.D.M. additionally provided primary MDS samples; K.J.H. conceived of and supervised the project, designed experiments, interpreted results, and wrote the manuscript; and all authors reviewed the manuscript.

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Footnotes

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Raw and processed omics data have been deposited, and are available to view on the European Genome-phenome Archive (EGA) (study ID: EGAS50000000946, data set ID: EGAD50000001389). Access for RIME proteomics data can be obtained and is available to view from the Mass-Spectrometry Interactive Virtual Environment (MassIVE) with the following link: <ftp://MSV000097058@massive.ucsd.edu>; username: MSV000097058; password: o8R048Rk2Z41HVwp.

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